

JAMF PRO

ADCS

Full Setup Guide

Ver. 1.0 · Active Directory Certificate Services · JAMF Pro Integration

CREATED BY
Ryan Adams

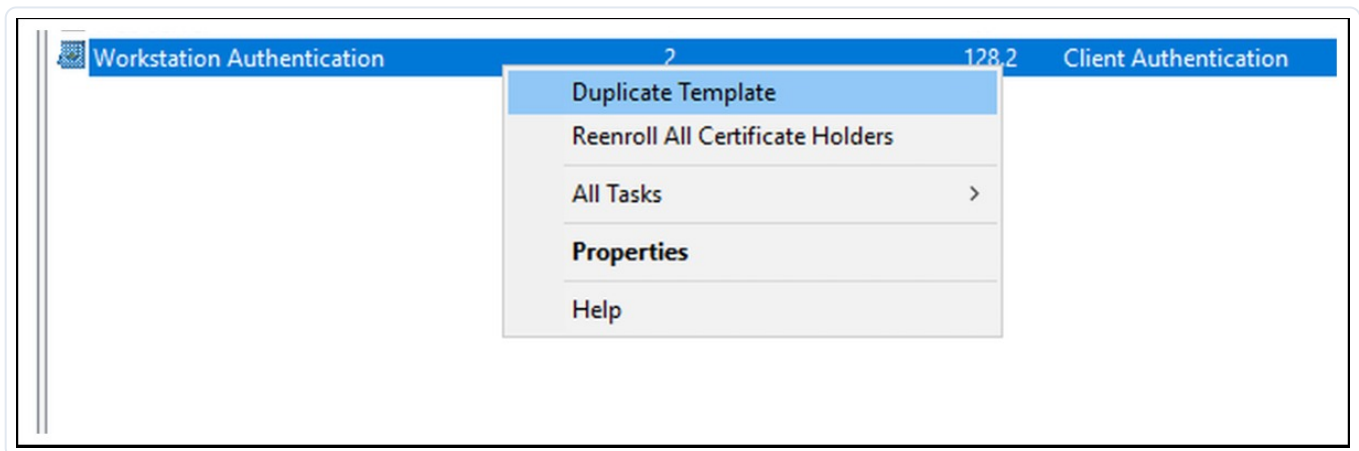
VERSION
Ver. 1.0

PLATFORM
JAMF Pro + Windows Server ADCS

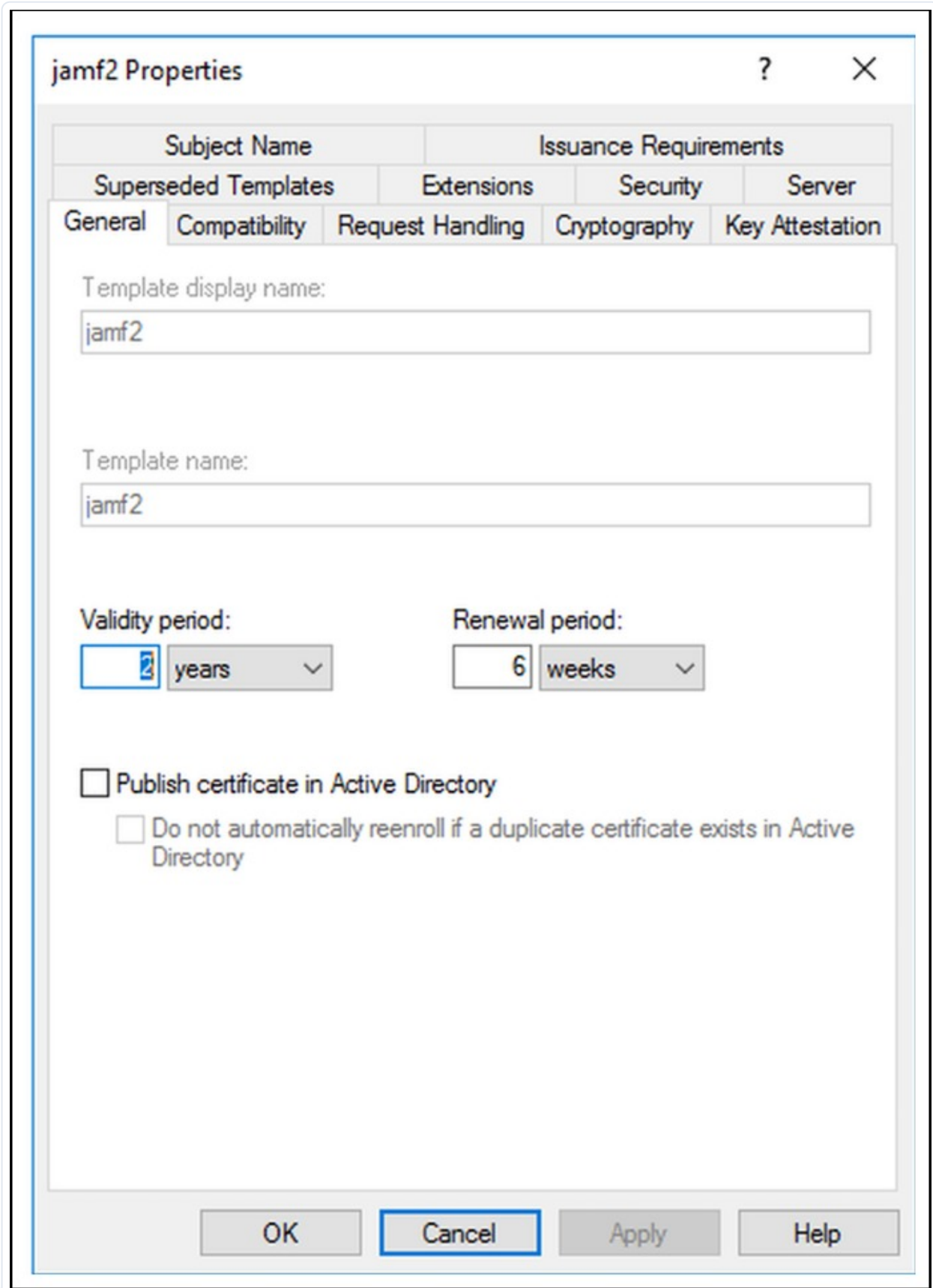
SECTIONS COVERED

- Internal CA
- General
- Compatibility
- Request Handling
- Cryptography
- Key Attestation
- Subject Name
- Issuance Requirements
- Superseded Templates
- Extensions
- Security
- Server
- Install ADCS Connector
- AD CS
- AD CS Server Integration
- JAMF Profile
- Network: Protocols
- Network: Trust
- Certificate

Internal CA



General



The image shows a Windows-style dialog box titled "jamf2 Properties". It has a standard title bar with a question mark and a close button. The dialog is divided into several tabs: "General", "Compatibility", "Request Handling", "Cryptography", "Key Attestation", "Superseded Templates", "Extensions", "Security", and "Server". The "General" tab is currently selected. Inside the "General" tab, there are two text input fields: "Template display name:" and "Template name:", both containing the text "jamf2". Below these fields are two groups of controls for validity and renewal periods. The "Validity period:" group has a numeric input field with the value "4", a unit dropdown menu set to "years", and a small icon of a calendar. The "Renewal period:" group has a numeric input field with the value "6", a unit dropdown menu set to "weeks", and a small icon of a calendar. At the bottom of the dialog, there are four buttons: "OK", "Cancel", "Apply", and "Help". The "Cancel" button is highlighted with a blue border.

jamf2 Properties

Subject Name Issuance Requirements

Superseded Templates Extensions Security Server

General Compatibility Request Handling Cryptography Key Attestation

Template display name:

jamf2

Template name:

jamf2

Validity period: Renewal period:

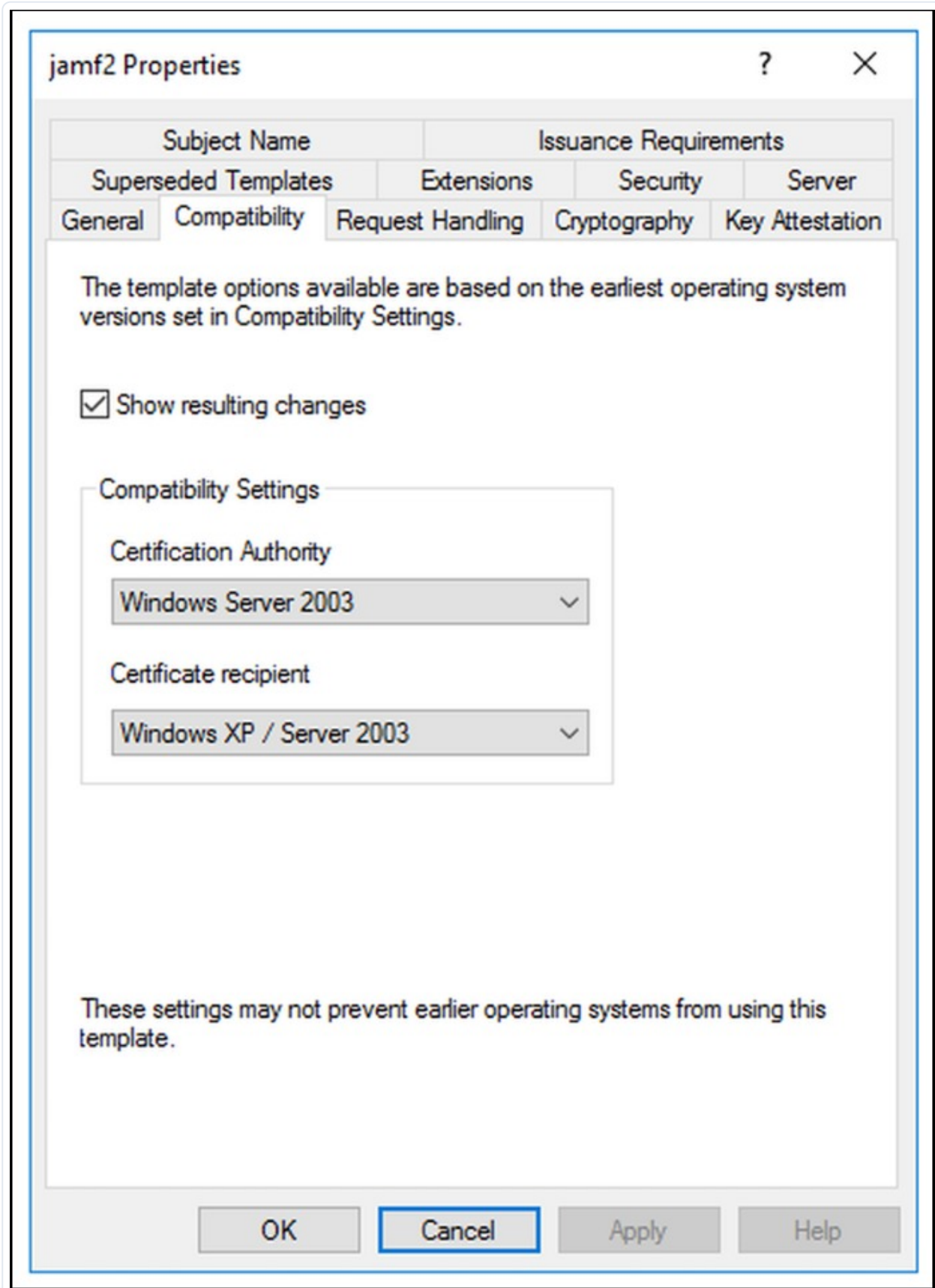
4 years 6 weeks

☐ Publish certificate in Active Directory

☐ Do not automatically reenroll if a duplicate certificate exists in Active Directory

OK Cancel Apply Help

Compatibility



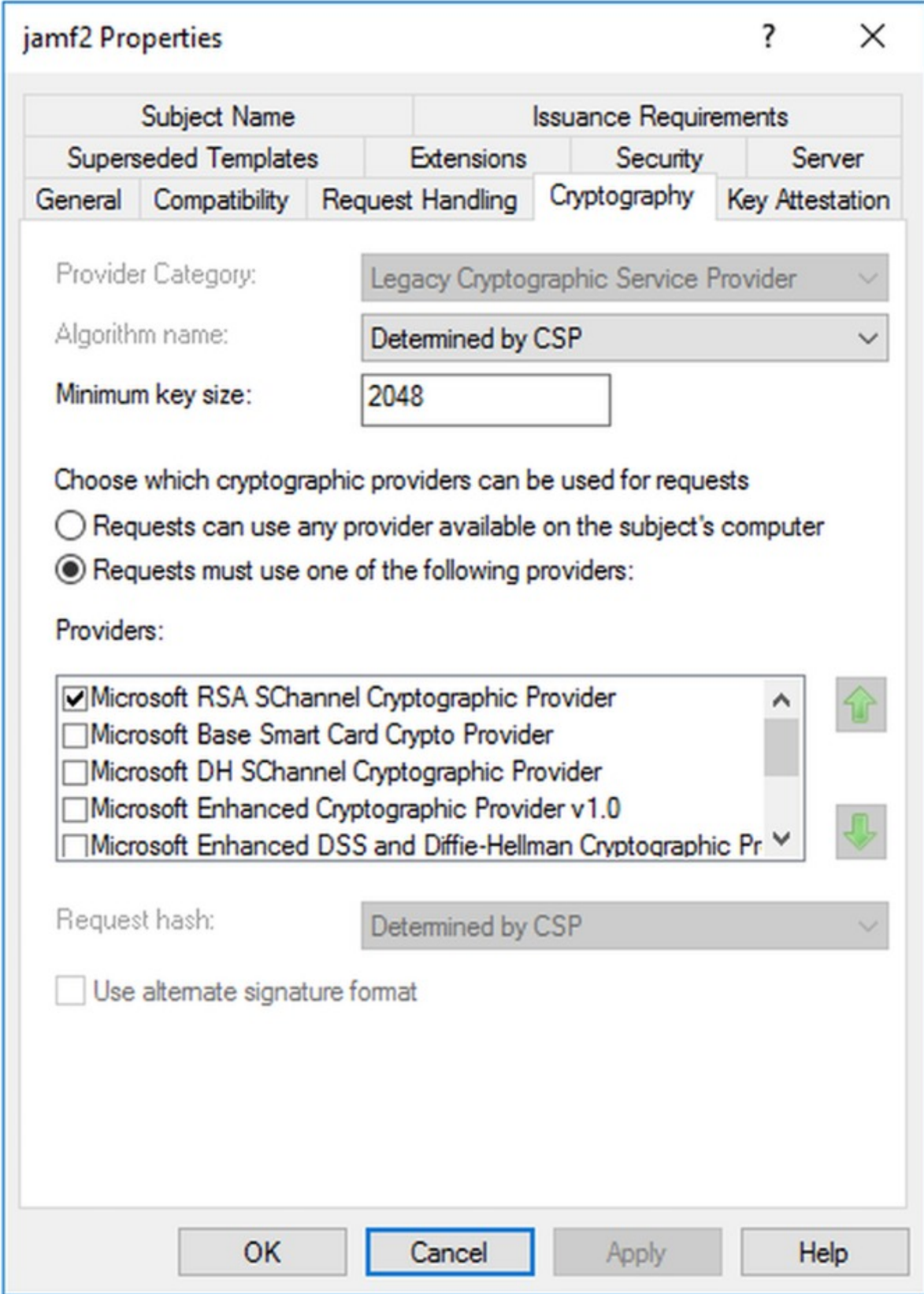
Request Handling

The screenshot shows the 'jamf2 Properties' dialog box with the 'Request Handling' tab selected. The dialog has a title bar with a question mark and a close button. Below the title bar is a tabbed interface with tabs for 'Subject Name', 'Issuance Requirements', 'Superseded Templates', 'Extensions', 'Security', 'Server', 'General', 'Compatibility', 'Request Handling' (selected), 'Cryptography', and 'Key Attestation'. The 'Request Handling' tab contains the following settings:

- Purpose:** A dropdown menu set to 'Signature and encryption'.
- ☐ Delete revoked or expired certificates (do not archive)
- ☐ Include symmetric algorithms allowed by the subject
- ☐ Archive subject's encryption private key
- ☐ Authorize additional service accounts to access the private key (*)
 Key Permissions...
- ☐ Allow private key to be exported
- ☐ Renew with the same key (*)
- ☐ For automatic renewal of smart card certificates, use the existing key if a new key cannot be created (*)
- Do the following when the subject is enrolled and when the private key associated with this certificate is used:**
 - ☒ Enroll subject without requiring any user input
 - ☐ Prompt the user during enrollment
 - ☐ Prompt the user during enrollment and require user input when the private key is used
- * Control is disabled due to [compatibility settings](#).

At the bottom of the dialog are four buttons: 'OK', 'Cancel' (highlighted with a blue border), 'Apply', and 'Help'.

Cryptography

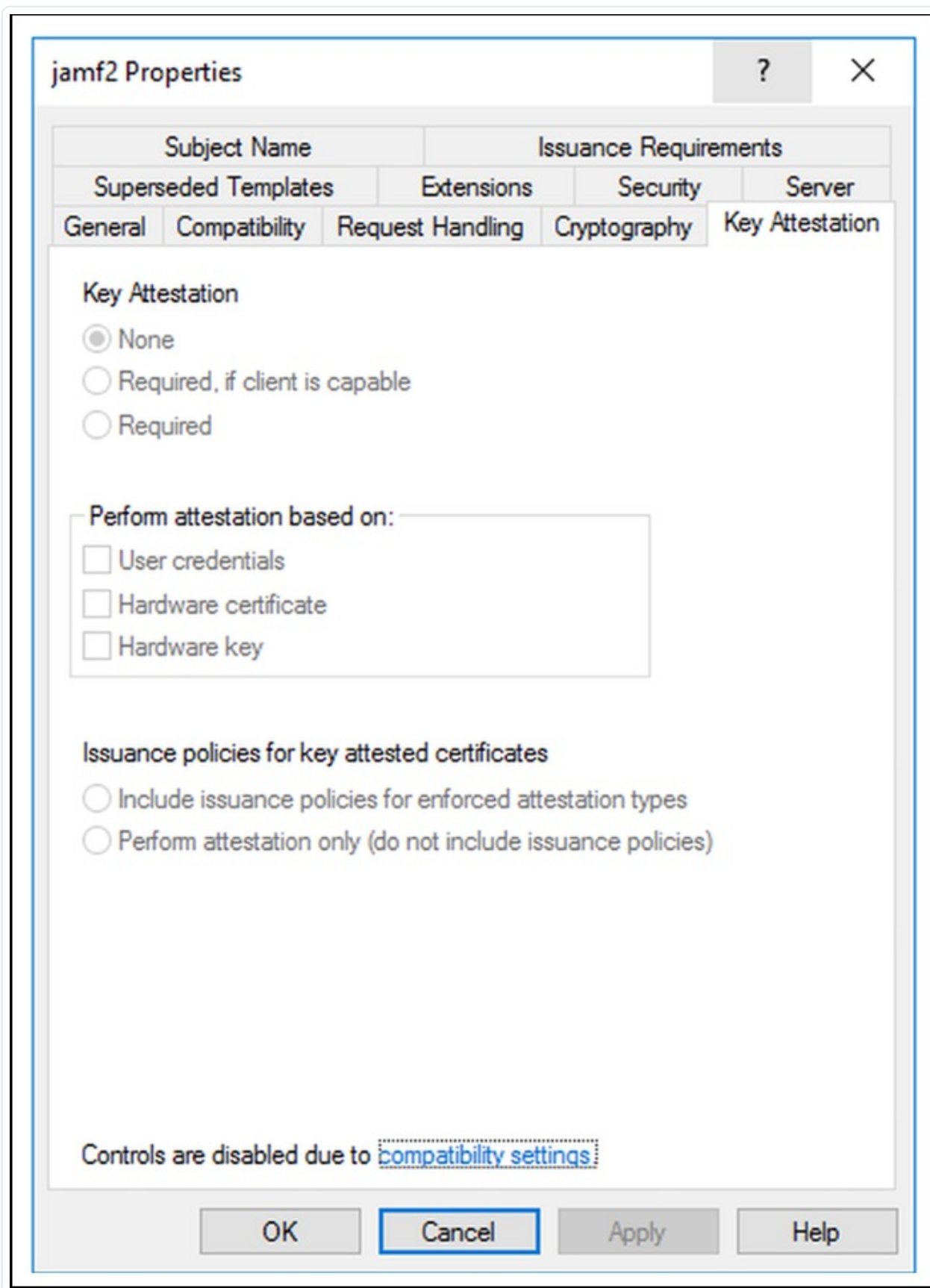


The image shows the 'jamf2 Properties' dialog box with the 'Cryptography' tab selected. The dialog has a title bar with a question mark and a close button. Below the title bar is a tabbed interface with the following tabs: 'Subject Name', 'Issuance Requirements', 'Superseded Templates', 'Extensions', 'Security', 'Server', 'General', 'Compatibility', 'Request Handling', 'Cryptography' (selected), and 'Key Attestation'. The 'Cryptography' tab contains the following settings:

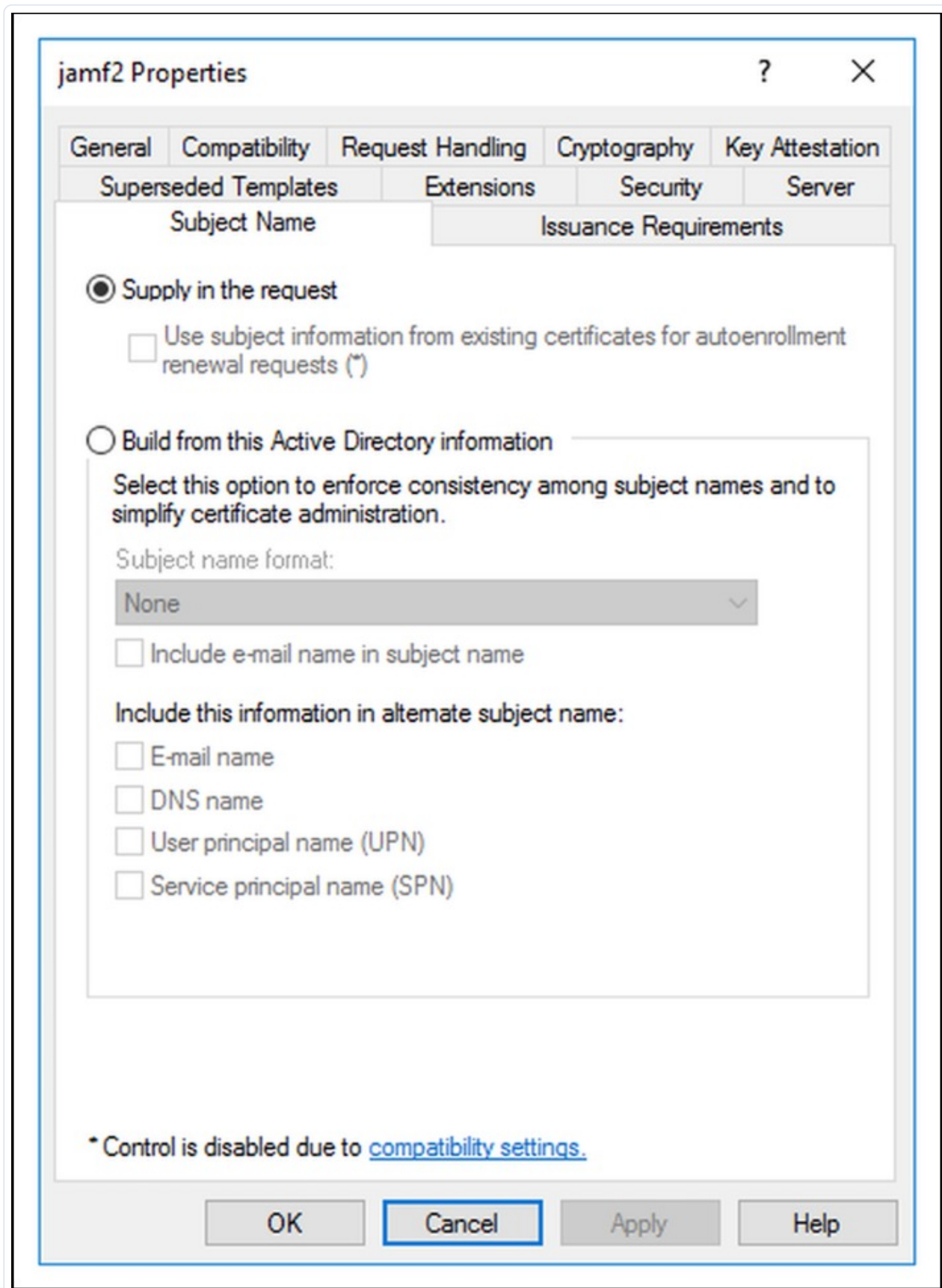
- Provider Category:** Legacy Cryptographic Service Provider (dropdown menu)
- Algorithm name:** Determined by CSP (dropdown menu)
- Minimum key size:** 2048 (text box)
- Choose which cryptographic providers can be used for requests:**
 - ☐ Requests can use any provider available on the subject's computer
 - ☒ Requests must use one of the following providers:
- Providers:** A list box containing the following providers:
 - ☒ Microsoft RSA SChannel Cryptographic Provider
 - ☐ Microsoft Base Smart Card Crypto Provider
 - ☐ Microsoft DH SChannel Cryptographic Provider
 - ☐ Microsoft Enhanced Cryptographic Provider v1.0
 - ☐ Microsoft Enhanced DSS and Diffie-Hellman Cryptographic PrThere are up and down arrow buttons to the right of the list box.
- Request hash:** Determined by CSP (dropdown menu)
- ☐ Use alternate signature format

At the bottom of the dialog are four buttons: 'OK', 'Cancel' (highlighted with a blue border), 'Apply', and 'Help'.

Key Attestation



Subject Name



The image shows the 'jamf2 Properties' dialog box with the 'Subject Name' tab selected. The dialog has a title bar with a question mark and a close button. Below the title bar are several tabs: 'General', 'Compatibility', 'Request Handling', 'Cryptography', 'Key Attestation', 'Superseded Templates', 'Extensions', 'Security', and 'Server'. The 'Subject Name' tab is active, showing options for how to supply subject information. The 'Supply in the request' option is selected with a radio button. Below it is a checkbox for 'Use subject information from existing certificates for autoenrollment renewal requests (*)'. The 'Build from this Active Directory information' option is unselected. Below it is a text box with the instruction 'Select this option to enforce consistency among subject names and to simplify certificate administration.' and a 'Subject name format:' dropdown menu currently set to 'None'. There is also a checkbox for 'Include e-mail name in subject name'. Further down, under the heading 'Include this information in alternate subject name:', there are four checkboxes: 'E-mail name', 'DNS name', 'User principal name (UPN)', and 'Service principal name (SPN)'. At the bottom of the dialog, there is a message: '* Control is disabled due to [compatibility settings](#).' and four buttons: 'OK', 'Cancel', 'Apply', and 'Help'.

jamf2 Properties

General Compatibility Request Handling Cryptography Key Attestation
Superseded Templates Extensions Security Server

Subject Name Issuance Requirements

☒ Supply in the request

☐ Use subject information from existing certificates for autoenrollment renewal requests (*)

☐ Build from this Active Directory information

Select this option to enforce consistency among subject names and to simplify certificate administration.

Subject name format:

None

☐ Include e-mail name in subject name

Include this information in alternate subject name:

☐ E-mail name

☐ DNS name

☐ User principal name (UPN)

☐ Service principal name (SPN)

* Control is disabled due to [compatibility settings](#).

OK Cancel Apply Help

Issuance Requirements

The screenshot shows the 'jamf2 Properties' dialog box with the 'Issuance Requirements' tab selected. The dialog has a title bar with a question mark and a close button. Below the title bar are several tabs: 'General', 'Compatibility', 'Request Handling', 'Cryptography', 'Key Attestation', 'Superseded Templates', 'Extensions', 'Security', and 'Server'. The 'Issuance Requirements' tab is active, showing settings for enrollment and reenrollment. The enrollment section includes checkboxes for 'CA certificate manager approval' and 'This number of authorized signatures' (set to 0), with a note that autoenrollment is not allowed if more than one signature is required. It also features dropdown menus for 'Policy type required in signature' and 'Application policy', and a list box for 'Issuance policies' with 'Add...' and 'Remove' buttons. The reenrollment section has radio buttons for 'Same criteria as for enrollment' (selected) and 'Valid existing certificate', with an unchecked checkbox for 'Allow key based renewal (*)'. A note states that subject information must be provided within the certificate request. A footer note indicates that a control is disabled due to compatibility settings. At the bottom are 'OK', 'Cancel', 'Apply', and 'Help' buttons.

jamf2 Properties

General Compatibility Request Handling Cryptography Key Attestation
Superseded Templates Extensions Security Server

Subject Name Issuance Requirements

Require the following for enrollment:

☐ CA certificate manager approval

☐ This number of authorized signatures: 0

If you require more than one signature, autoenrollment is not allowed.

Policy type required in signature:

Application policy:

Issuance policies:

Add...

Remove

Require the following for reenrollment:

☒ Same criteria as for enrollment

☐ Valid existing certificate

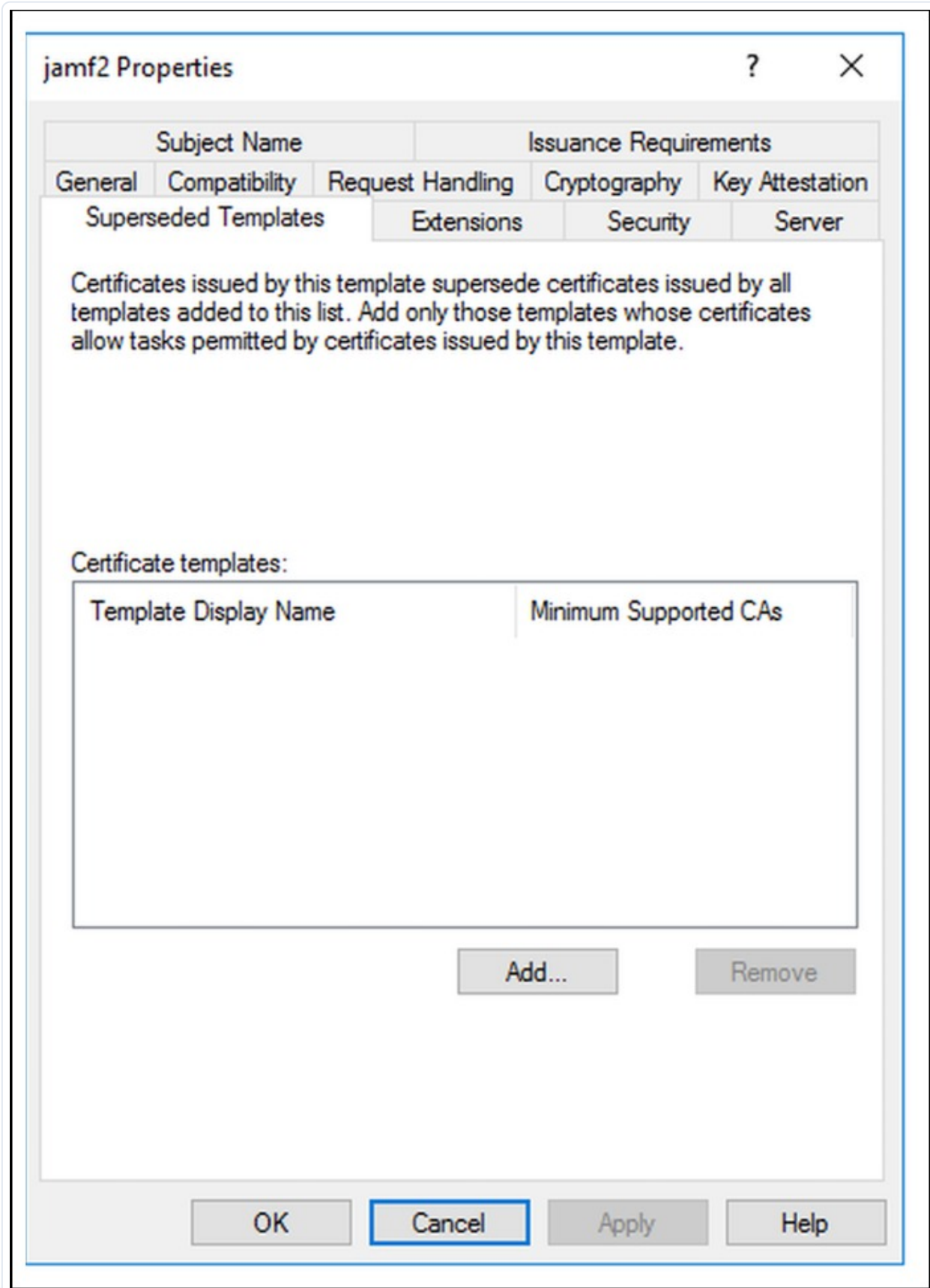
☐ Allow key based renewal (*)

Requires subject information to be provided within the certificate request.

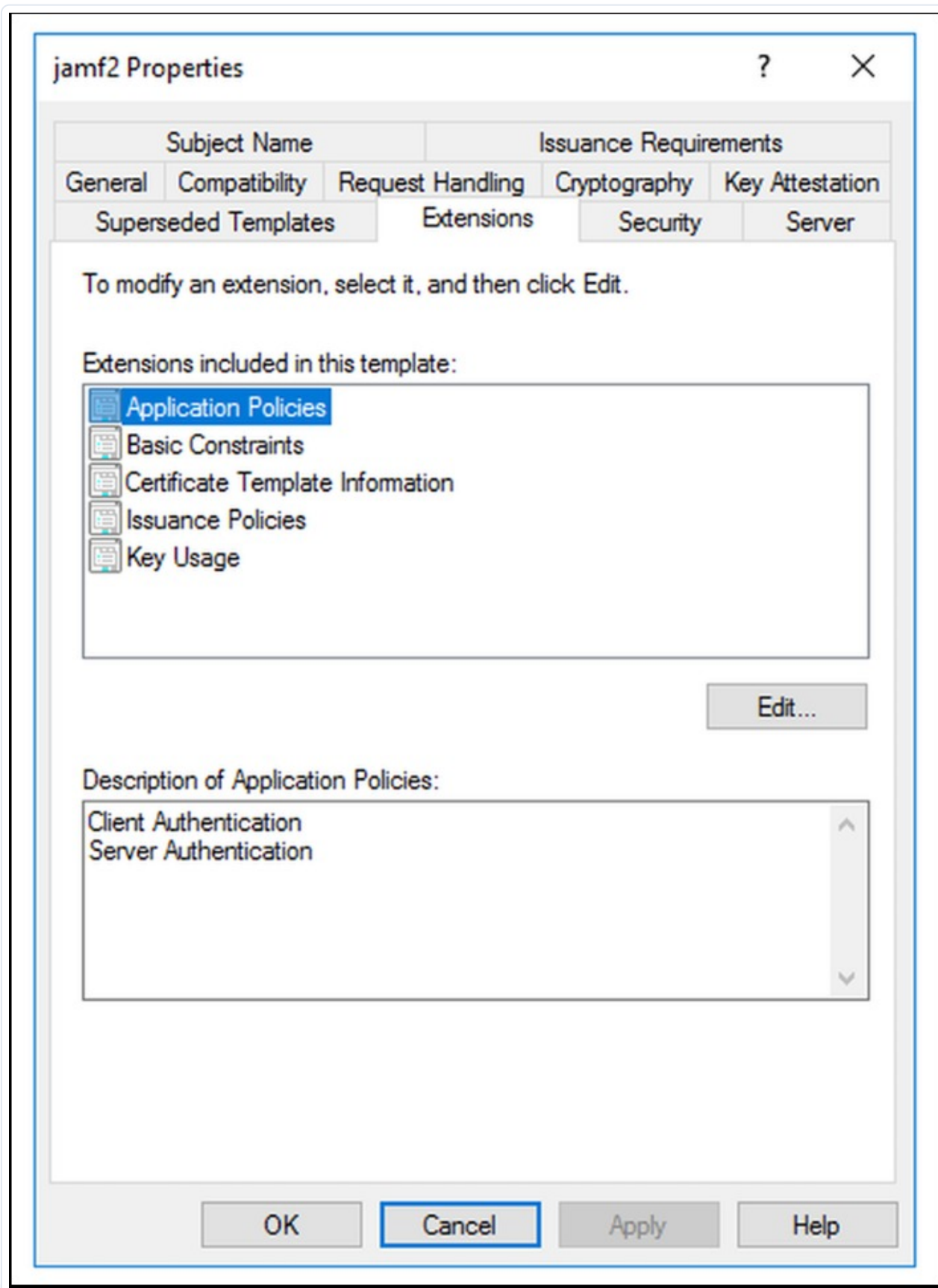
* Control is disabled due to [compatibility settings](#).

OK Cancel Apply Help

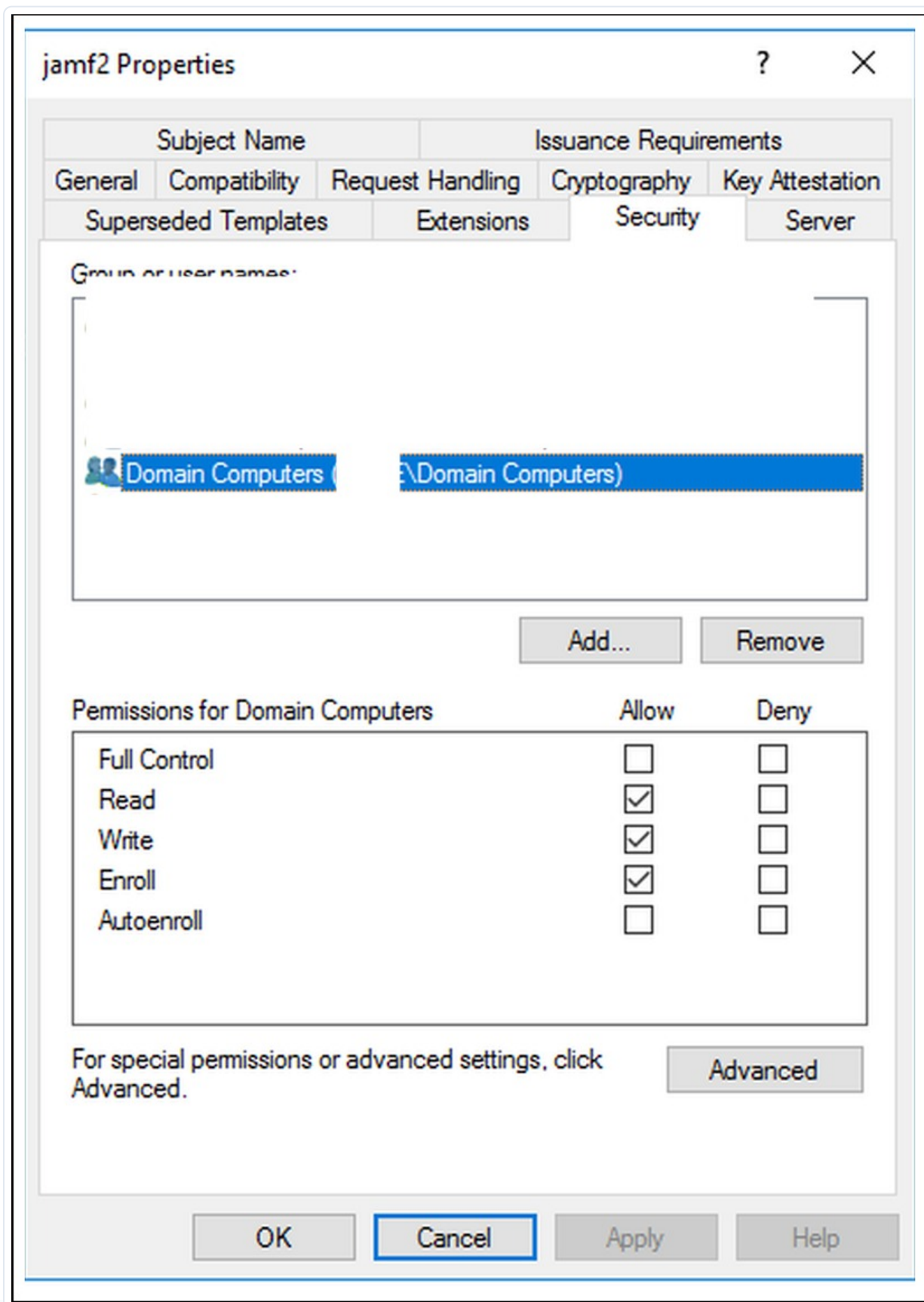
Superseded Templates



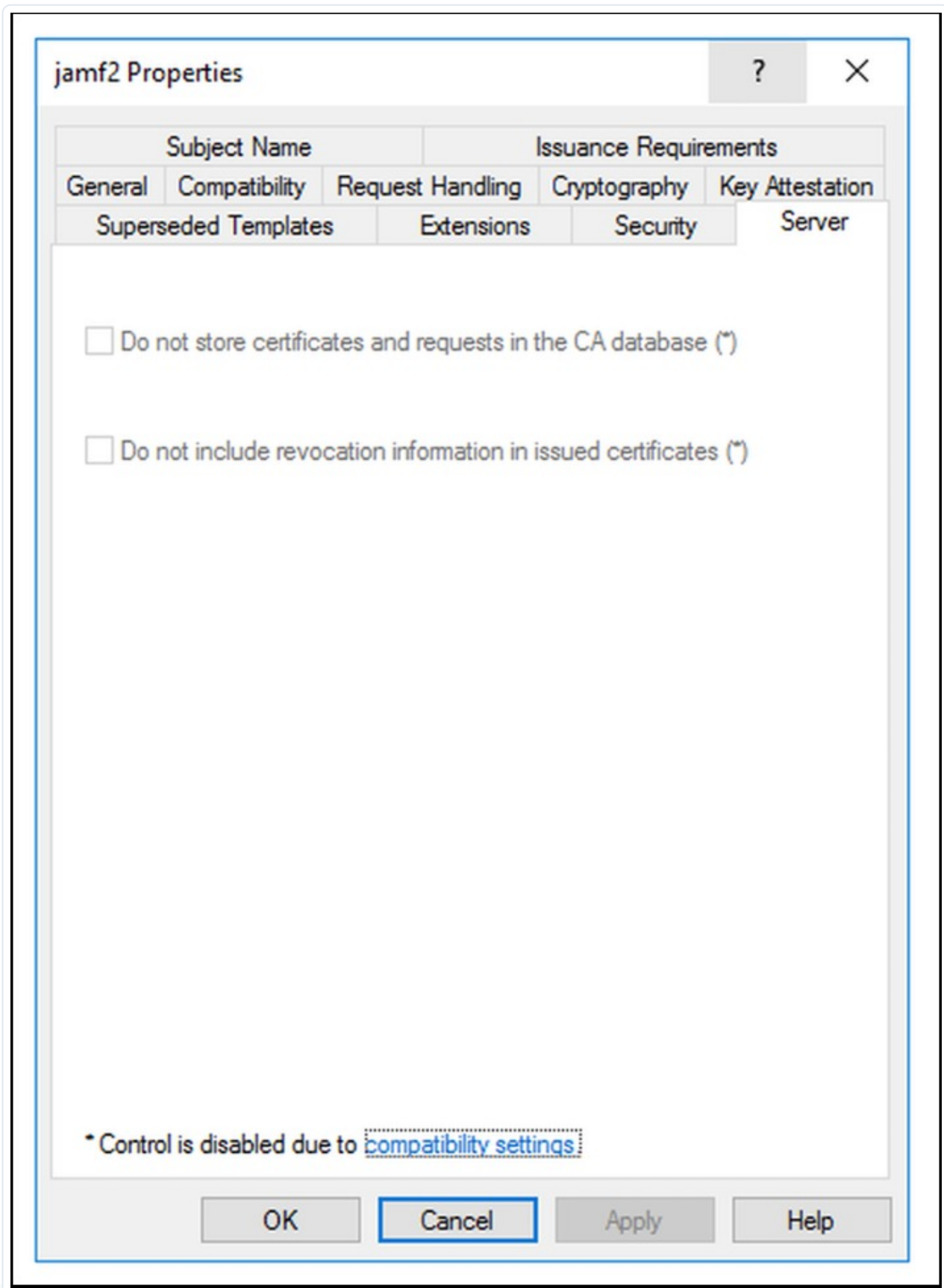
Extensions



Security



Server

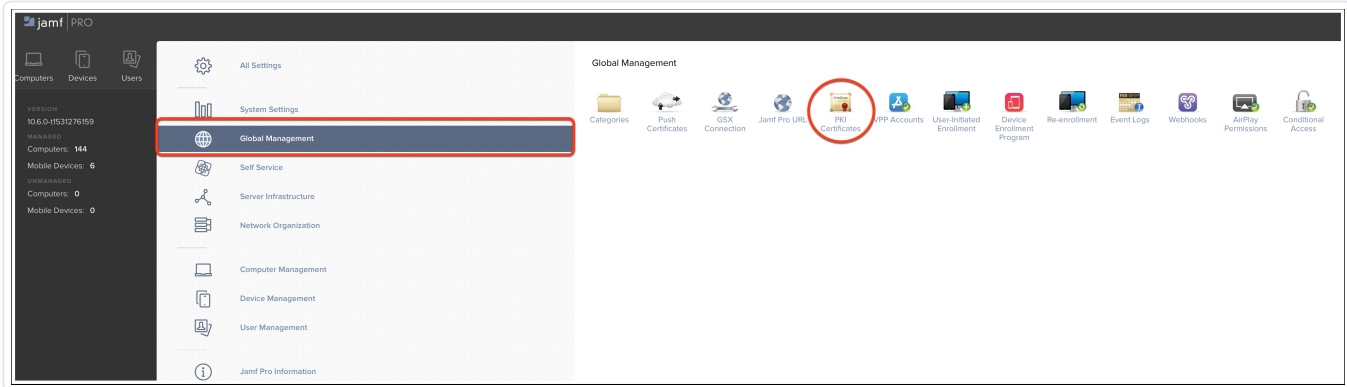


Install ADCS Connector

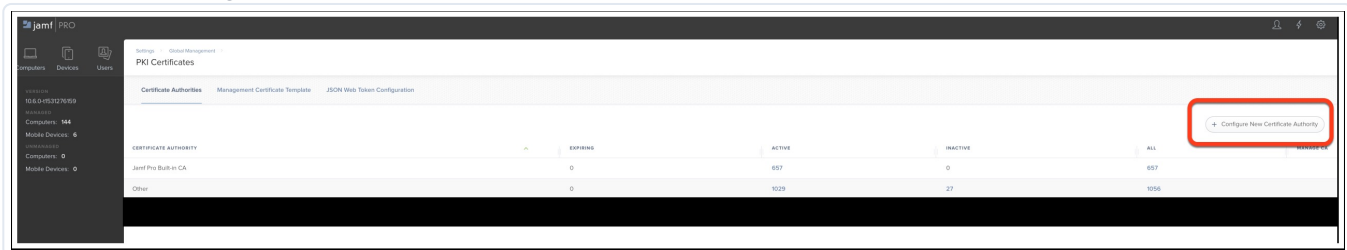
JAMF PKI Setup

```
Untitled1 - Notepad
File Edit Format View Help
.\deploy.ps1 -fqdn public-dns-name.mydomain.com -jamfProDn my-JAMF-PRO-Cloud-URL.jamfcloud.com -cleanInstall
```

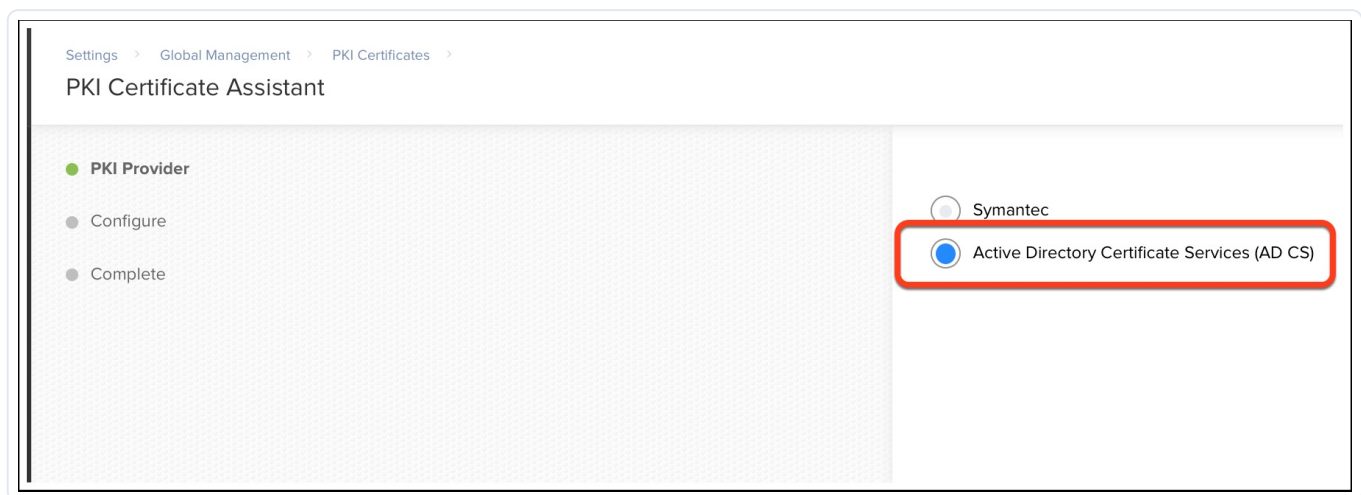
Configure New Certificate Authority



ADCS Connector Configuration



AD CS



AD CS Server Integration

AD CS Server Integration Settings to integrate Jamf Pro with Active Directory Certificate Service

FULLY QUALIFIED DOMAIN NAME Fully qualified domain name of the certificate authority server

CA NAME Name of the certificate authority

Jamf AD CS Connector Settings for Jamf Pro to request a certificate from Active Directory Certificate Service via the Jamf AD CS Connector

URL OR IP ADDRESS URL or IP address of the Jamf AD CS Connector

SERVER CERTIFICATE Certificate required for Jamf Pro to verify identity of the Jamf AD CS Connector (.pem, .cer)

CLIENT CERTIFICATE Certificate required for Jamf Pro to authenticate with the Jamf AD CS Connector (.pfx, .p12)

JAMF Profile

Corporate

OptionsScope

GeneralNetwork (Physical Configured)Certificate (Physical Configured)

Wi-Fi

SERVICE SET IDENTIFIER (SSID) Identification of the wireless network to connect to.
Corporate

Hidden Network

Auto Join

PROXY SETUP Configure proxy settings to be used with this network.
Name

SECURITY TYPE Wireless network encryption to use when connecting.
WPA2 Enterprise

Use as a Login Window configuration

Network Security Settings Configurations options for 802.11i network authentication.

Protocols	Trust
ACCEPTED EAP TYPES Authentication protocols supported on target network	
TLS TTLS LEAP PEAP EAP-FAST EAP-SIM EAP-AKA	

USERNAME Username for connection to the network

TLS MINIMUM VERSION
Name

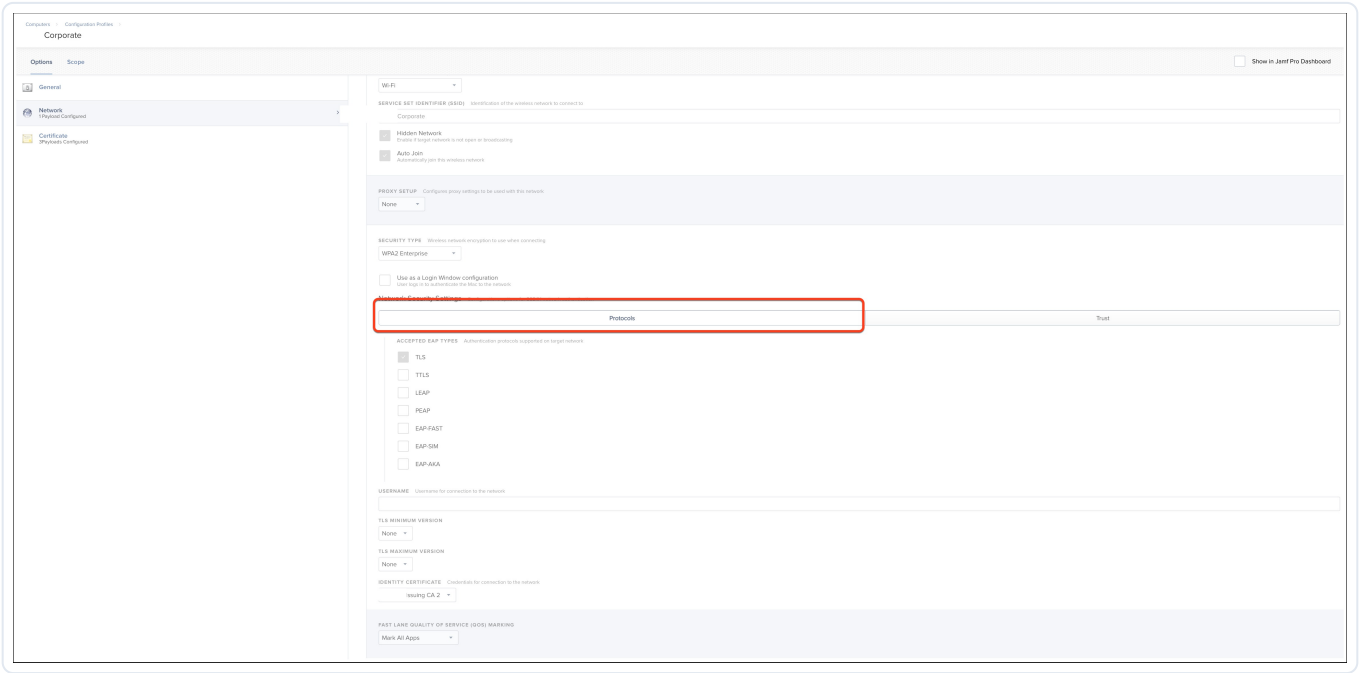
TLS MAXIMUM VERSION
Name

IDENTITY CERTIFICATE Certificate for connection to the network
Issuing CA 2

POST LANE QUALITY OF SERVICE (QoS) MARKING
Mark All Apps

Show in Jamf Pro Dashboard

Network: Protocols



Network: Trust

The screenshot displays the Jamf Pro configuration interface for a Corporate profile. The left sidebar shows the navigation menu with 'Network' selected. The main content area is titled 'Network' and contains several sections: 'NETWORK INTERFACE' (set to Wi-Fi), 'SERVICE SET IDENTIFIER (SSID)' (set to Corporate), 'PROXY SETUP' (set to None), 'SECURITY TYPE' (set to WPA2 Enterprise), and 'Network Security Settings'. The 'Network Security Settings' section is expanded, showing 'Protocols' and 'Trust' tabs. The 'Trust' tab is selected and highlighted with a red box. Below this, the 'IDENTITY CERTIFICATE' section is expanded, showing 'Trusted Certificates' and 'Trusted Server Certificate Names'. The 'Trusted Certificates' section is also highlighted with a red box, showing a list of certificates: 'Issuing CA' and 'Root CA'. The 'Trusted Server Certificate Names' section shows a list of certificates: 'BAD63a' and 'BAD64a'. The 'Certificate Common Name' section shows a list of certificates: 'BAD63a' and 'BAD64a'. The 'Allow Trust Exceptions' checkbox is checked. The 'Root CA' checkbox is checked. The 'Issuing CA' checkbox is checked. The 'Trusted Server Certificate Names' section shows a list of certificates: 'BAD63a' and 'BAD64a'. The 'Certificate Common Name' section shows a list of certificates: 'BAD63a' and 'BAD64a'. The 'Allow Trust Exceptions' checkbox is checked. The 'Root CA' checkbox is checked. The 'Issuing CA' checkbox is checked.

Corporate

Options Scope

General

Network (Profile Configured)

Certificate (Profile Configured)

Network

NETWORK INTERFACE: Wi-Fi

SERVICE SET IDENTIFIER (SSID): Corporate

PROXY SETUP: None

SECURITY TYPE: WPA2 Enterprise

Use as a Login Window configuration: ☐

Network Security Settings: Protocols Trust

USERNAME: (empty)

TLS MINIMUM VERSION: None

TLS MAXIMUM VERSION: None

IDENTITY CERTIFICATE: Issuing CA 2

TRUSTED CERTIFICATES: Issuing Cert, Root CA

TRUSTED SERVER CERTIFICATE NAMES: (empty)

CERTIFICATE COMMON NAME: BAD63a, BAD64a

Allow Trust Exceptions: ☒

Root CA: ☒

Issuing CA: ☒

Root Line Quality of Service (QoS) Marking: Mark All Apps

Certificate

Corporate

Options Scope

General

Network
1 Payload Configured

Certificate
3 Payloads Configured

▼ Certificate

CERTIFICATE NAME Display name of the certificate credential
Issuing CA 2

CERTIFICATE SUBJECT Representation of a X.500 name (e.g. O=CompanyName, CN=FOO)
CN=\$USERNAME@ti .com

SUBJECT ALTERNATIVE NAME TYPE The type of a subject alternative name
None

TEMPLATE NAME The name of the certificate template, usually Machine or User
jamf2

▼ Certificate

CERTIFICATE NAME Display name of the certificate credential
Issuing Cert

CERTIFICATE
FILENAME
Issuing-2.cer

ISSUER NAME
CN=Ti Root CA

EXPIRATION DATE
03/20/2037

PASSWORD Password used to secure certificate credentials

VERIFY PASSWORD

▼ Certificate

CERTIFICATE NAME Display name of the certificate credential
Root CA

CERTIFICATE
FILENAME
Root.cer

ISSUER NAME
CN=Ti Root CA

EXPIRATION DATE
03/20/2057